

Algorithms analysis and Design

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Chapter 2

Fundamentals of the Analysis of Algorithm Efficiency

OUTLINE

- ▶ **The Analysis Framework**
- ▶ **Asymptotic Notations & Basic Efficiency Classes**
- ▶ **Analysis of Non-recursive Algorithms**
- ▶ **Analysis of Recursive Algorithms**

Examples of Algorithm analysis

" $f(n)$ is $O(g(n))$ " : $f(n) \leq c(g(n))$ for $n \geq n_0$

► Constant-Time Operations

All of the primitive operations are **assumed to run in constant time $O(1)$**

► Finding the Maximum of an Array $O(n)$

```

1  /** Returns the maximum value of a nonempty array of numbers. */
2  public static double arrayMax(double[] data) {
3      int n = data.length;
4      double currentMax = data[0];
5      for (int j=1; j < n; j++)
6          if (data[j] > currentMax)
7              currentMax = data[j];
8      return currentMax;
9  }
```

Annotations for the code above:

- Line 3: 1 (constant time)
- Line 4: 1 (constant time)
- Line 5: n (loop iterations)
- Line 6: $n-1$ (loop iterations)
- Line 7: $n-1$ (loop iterations)
- Line 8: 1 (constant time)

Summary of operations:

- $3n + 1$ (total operations)
- $4n \leq cn$ (simplified bound)
- $c = 4, n_0 \geq 1$ (constants)

$$3n + 1 \leq c g(n)$$

Examples of Algorithm analysis

► Element Uniqueness

$O(n^2)$

$$n^2 + 2n + 3 \leq c \cdot g(n)$$
$$7n^2 \leq c \cdot n^2$$

```
1  /** Returns true if there are no duplicate elements in the array. */
2  public static boolean unique1(int[ ] data) {
3      int n = data.length;
4      for (int j=0; j < n-1; j++)
5          for (int k=j+1; k < n; k++)
6              if (data[j] == data[k])
7                  return false;
8      return true;
9  }
```

$n^2 + 2n + 3 \leq c \cdot g(n)$

Examples of Algorithm analysis

► Prefix Averages (Quadratic) $O(n^2)$

```
1  /** Returns an array a such that, for all j, a[j] equals the average of x[0], ..., x[j]. */
2  public static double[] prefixAverage1(double[] x) {
3      int n = x.length;
4      double[] a = new double[n];           // filled with zeros by default
5      for (int j=0; j < n; j++) {
6          double total = 0;                 // begin computing x[0] + ... + x[j]
7          for (int i=0; i <= j; i++)
8              total += x[i];
9          a[j] = total / (j+1);             // record the average
10     }
11     return a;
12 }
```



Examples of Algorithm analysis

Example 1 $O(1)$

`int x= n + 12;` ←----- 1

Example 2 $O(n)$

`for(int i=0;i<n;i++) {` ←----- n
 ..
 ..
}



Examples of Algorithm analysis

Example 3 $O(n)$

```
for(int i = 0; i < n; i+=2) ←-----  $n / 2$   
    sum++;
```

Example 4 $O(nm)$

```
for(int i=0;i<n;i++){ ←-----  $n$   
    for(int i=0;i<m;i++){ ←-----  $n \times m$   
        ..  
    }  
}
```


Examples of Algorithm analysis

Example 5 $O(n)$

for(int i = 0; i < n; i+=2) ← $n / 2$

sum++;

for(int j = 0; j < n; j++) ← n

sum++;

Example 6 $O(n^2)$

for(int i = 0; i < n; i++) ← n

for(int j = 0; j < n; j++) ← $n \times n$

sum++;

Examples of Algorithm analysis

Example 7 $O(n^2)$

```
for(int i = 0; i < n; i++)  
    for( int j = 0; j < i; j++)  
        sum++;
```

Example 8 $O(n^3)$

```
for(int i = 0; i < n; i++)  
    for( int j = 0; j < n * n; j++)  
        sum++;
```

Examples of Algorithm analysis

Example 9 $O(n^5)$

```
for(int i = 0; i < n; i++)
```

```
    for( int j = 0; j < n * n; j++)
```

```
        for(int k = 0; k < j; k++)
```

```
            sum++;
```

----- $(n^2(n^2 + 1)/2) \times n$

Example 10 $O(\log(n))$

```
for(int i = 1; i < n; i = i * 2)
```

----- $k = \log(n)$

```
    sum++;
```

$i = 1, i = 2, i = 4, \dots, i = n$

$i = 2^0, i = 2^1, i = 2^2, \dots, i = 2^k$

$2^k = n$

Examples of Algorithm analysis

Example 11 $O(\log(n))$

```
while(n>1){ ←-----  $k = \log(n)$ 
    n=n/2;}
```

$n, n/2, n/4, \dots, n/n = 1$

$n/2^0, n/2^1, n/2^2, \dots, (n/2^k) = 1$

$2^k = n$

Example 12 $O(n^2 \log n)$

```
for(c = 0; c < n; c++){
    for(i = 0; i < n; i++){
        for(x = 0; x < n; x = x*2){
            a+=1; }
        }
    }
```

Examples of Algorithm analysis

Example 13 $O(n)$

```
for (int i = 0; i < 22; i++)  
    for (int j = 0; j < n; j++)  
        sum++;
```

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